SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Science, Mathematics and Technology

SSILP Math

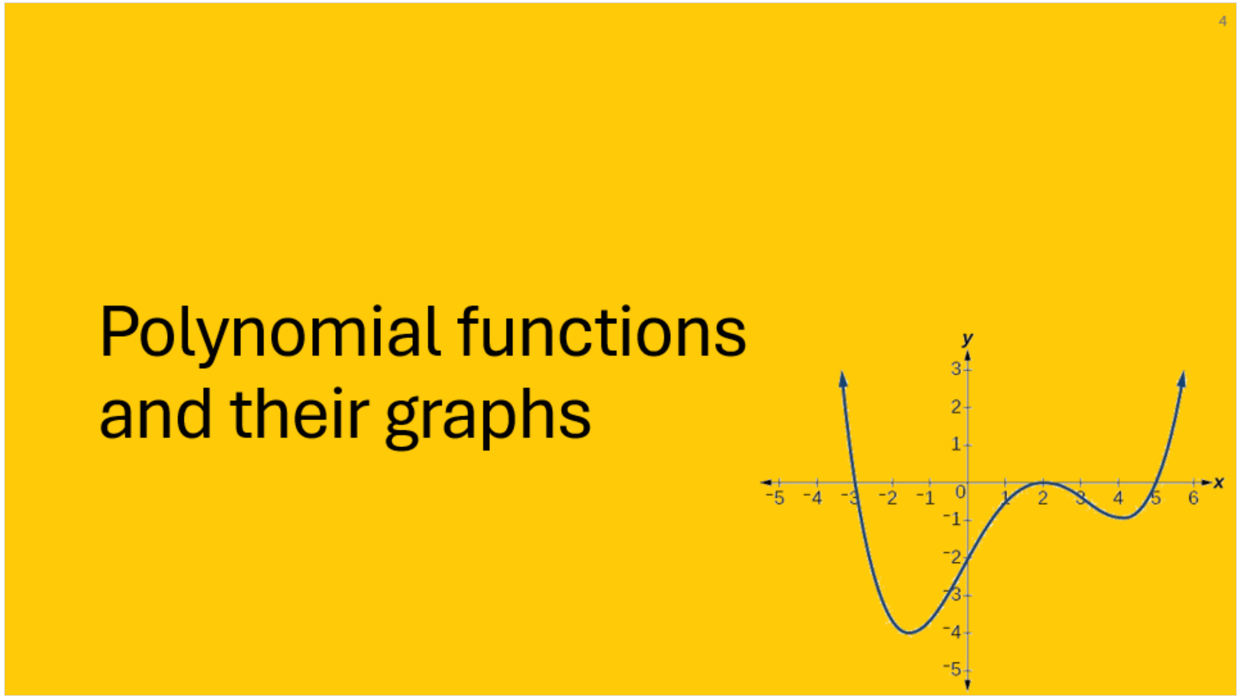
Beginner

POLYNOMIAL FUNCTIONS & THEIR GRAPHS

Learning objectives

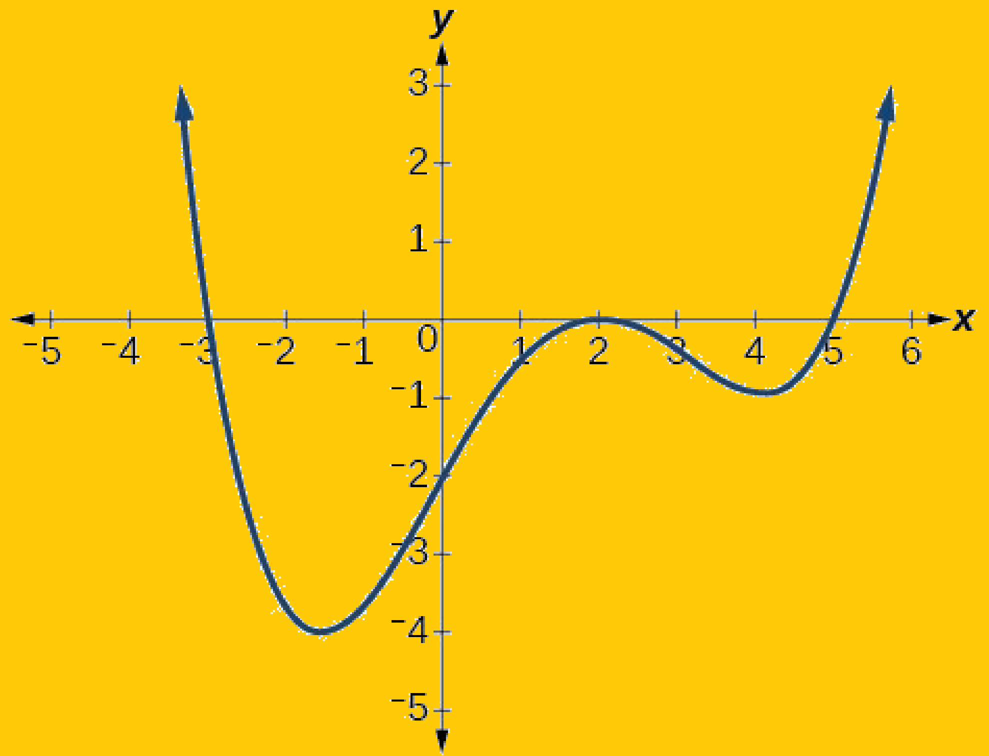
- Understand polynomial functions and their graphs
- Graph basic polynomial functions and understand their general shapes for different degrees
- Identify the real zeroes and end behavior of polynomial functions
- Use the Intermediate Value Theorem to graph polynomial functions.
- Graph polynomial functions using their zeros and test points to determine the sign of the polynomial in intervals.
- Understand the concept of multiplicities and how they affect the shape of the graph near a zero.

Session outline



*Contd from previous session

Polynomial functions and their graphs



Overview

What are polynomial functions?

How to graph basic polynomial functions

- What is the end behaviour?
- What are the x-intercepts/ zeros?
- What is the shape of the graph near the x-intercept/ zero?

Local Max and min of Polynomials

Polynomial functions: Formal definition

POLYNOMIAL FUNCTIONS

A **polynomial function of degree** n is a function of the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where n is a nonnegative integer and $a_n \neq 0$.

The numbers $a_0, a_1, a_2, \ldots, a_n$ are called the **coefficients** of the polynomial.

The number a_0 is the **constant coefficient** or **constant term**.

The number a_n, the coefficient of the highest power, is the **leading coefficient**, and the term $a_n x^n$ is the **leading term**.

Leading coefficient 3

Degree 5

Constant term -6

$$3x^5 + 6x^4 - 2x^3 + x^2 + 7x - 6$$

Leading term $3x^5$

Coefficients 3, 6, -2, 1, 7, and -6

Identifying parts of a polynomial

Polynomial	Degree	Leading term	Constant term
$P(x) = 4x - 7$	1	$4x$	-7
$P(x) = x^2 + x$	2	x^2	0
$P(x) = 2x^3 - 6x^2 + 10$	3	$2x^3$	10
$P(x) = -5x^4 + x - 2$	4	$-5x^4$	-2

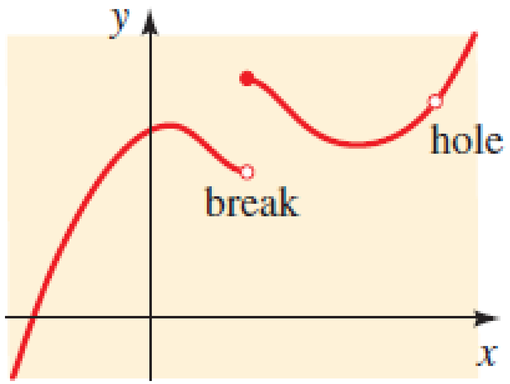
Characteristics of Graphs of polynomials

The graph of a polynomial function is **continuous.**

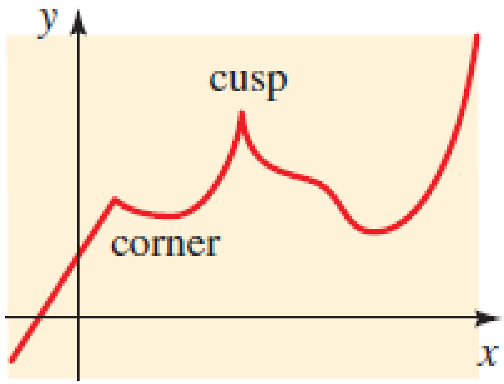
→ It has **no breaks** or **holes.**

The graph of a polynomial function is a **smooth curve.**

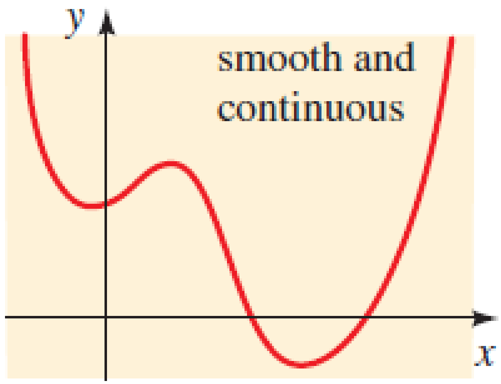
→it has **no corners** or **sham points (cusps).**



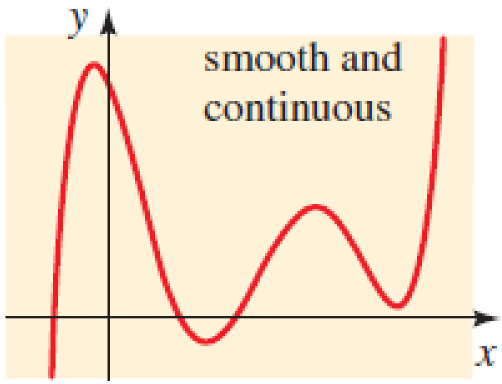
Not the graph of a polynomial function



Not the graph of a polynomial function



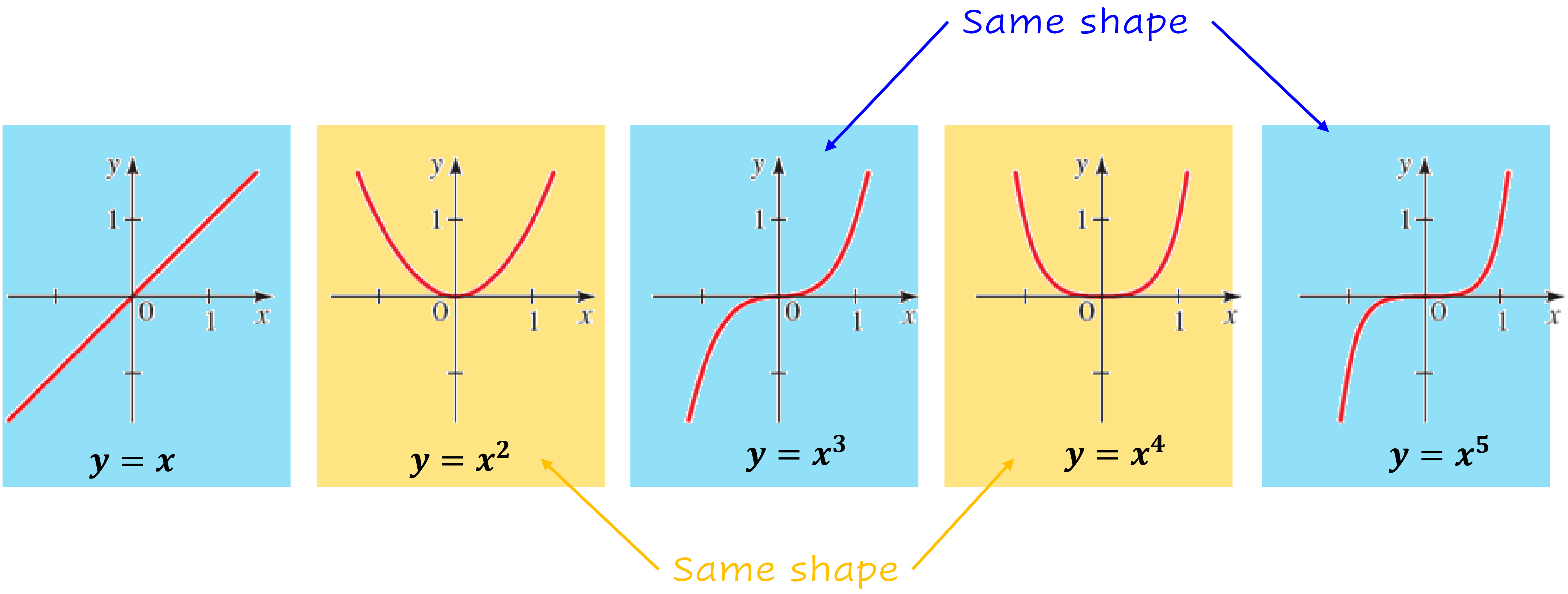
Graph of a polynomial function



Graph of a polynomial function

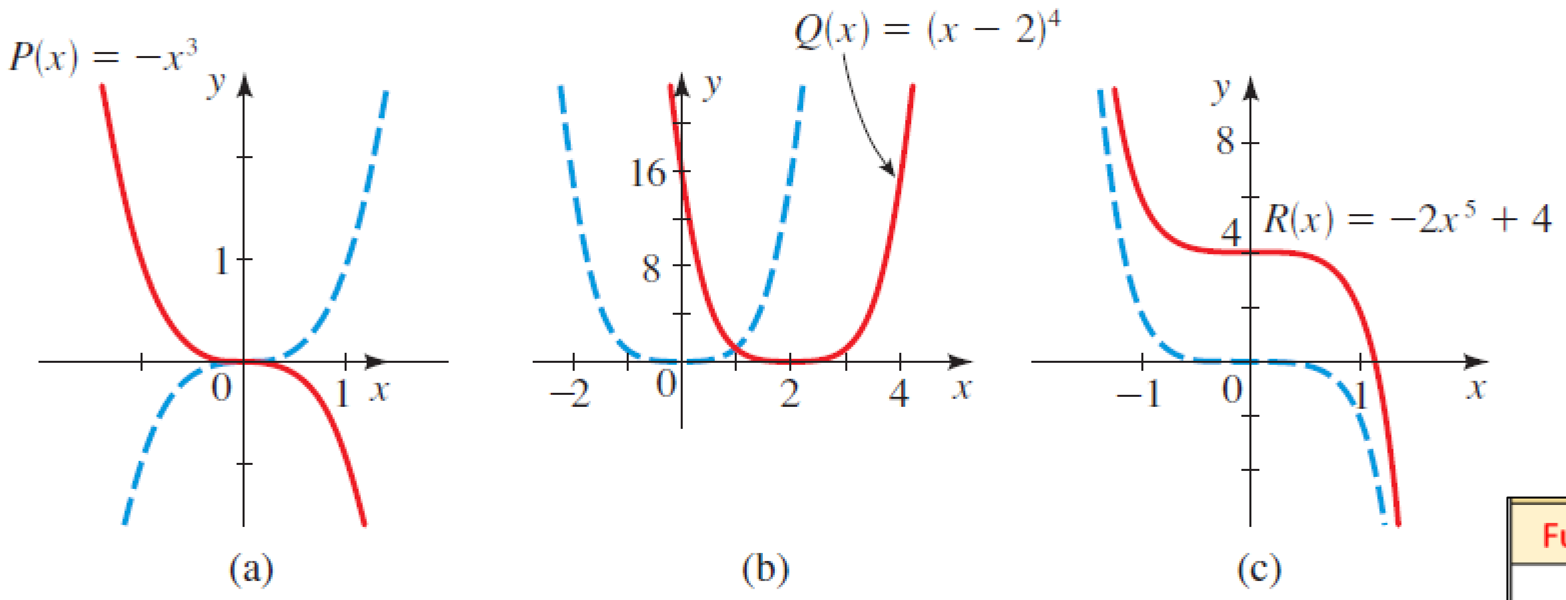
Graphing basic polynomial functions

The simplest polynomial functions are the monomials $P(x) = x^n$



The graph of $P(x) = x^n$ has
the **same general shape** as the graph of $y = x^2$ when **n is even**
And
the **same general shape** as the graph of $y = x^3$ when **n is odd**.

Transformation rules recap



Observe the parent graphs (in blue)

How are the graphs being transformed in each case?

Function Notation	Type of Transformation
f(x) + d	Vertical translation up d units
f(x) – d	Vertical translation down d units
f(x + c)	Horizontal translation left c units
f(x – c)	Horizontal translation right c units
–f(x)	Reflection over x-axis
f(–x)	Reflection over y-axis
af(x)	Vertical stretch for $\|a\| > 1$
	Vertical compression for $0 < \|a\| < 1$
f(bx)	Horizontal compression for $\|b\| > 1$
	Horizontal stretch for $0 < \|b\| < 1$

Graphing polynomials: End behaviour

The end behavior of a polynomial is a description of
what happens as **x becomes large** in the **positive or negative direction.**

Notation for the end behaviour

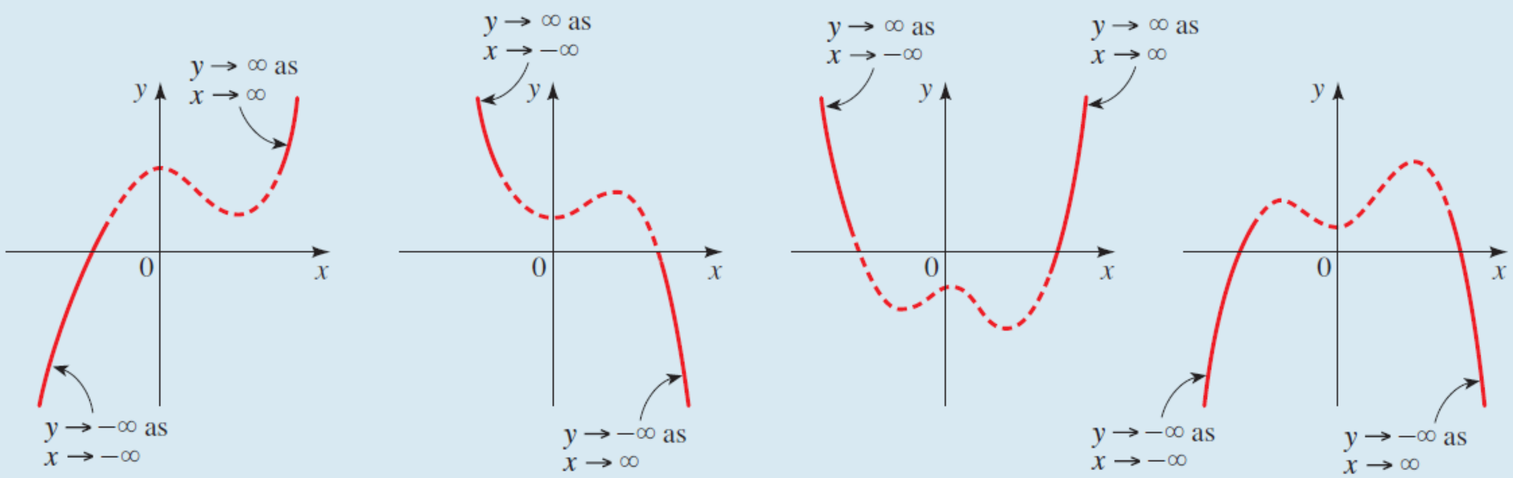
Symbol	Meaning
$x \to \infty$	x goes to infinity; that is, x increases without bound
$x \to -\infty$	x goes to negative infinity; that is, x decreases without bound

Four possibilities for end behaviour of polynomials

END BEHAVIOR OF POLYNOMIALS

The end behavior of the polynomial $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ is determined by the degree n and the sign of the leading coefficient a_n, as indicated in the following graphs.

***P* has odd degree** | ***P* has even degree**



Leading coefficient positive | Leading coefficient negative | Leading coefficient positive | Leading coefficient negative

Example: End behavior of a polynomial

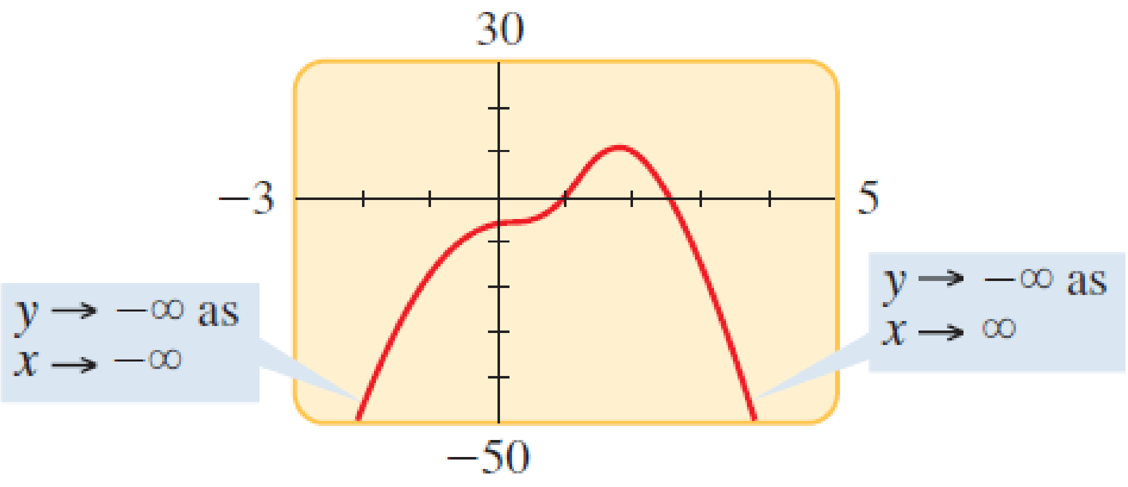
Determine the end behavior of the polynomial

$$P(x) = -2x^4 + 5x^3 + 4x - 7$$

SOLUTION The polynomial P has degree 4 and leading coefficient -2. Thus P has *even* degree and *negative* leading coefficient, so it has the following end behavior.

$$y \to -\infty \quad \text{as} \quad x \to \infty \qquad \text{and} \qquad y \to -\infty \quad \text{as} \quad x \to -\infty$$

The graph in Figure 4 illustrates the end behavior of P.



EXAMPLE

1.1 End behaviour

(i) Describe the end behavior of each polynomial function.
(ii) Match the polynomial function with one of the graphs I–VI.

a $P(x) = x(x^2 - 4)$

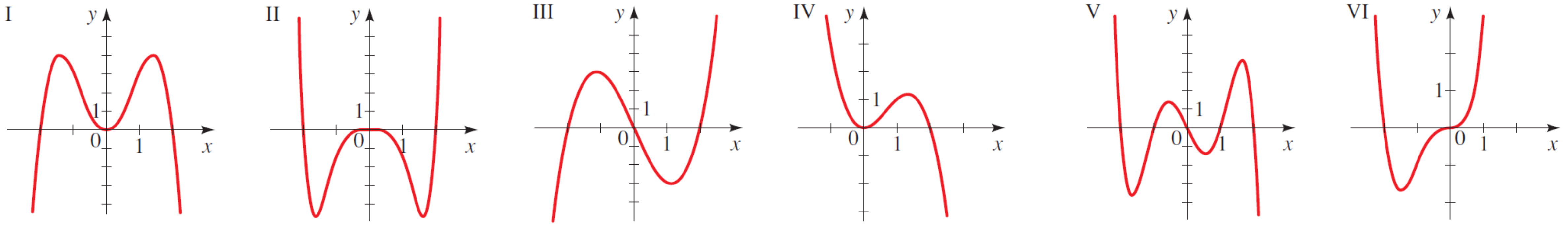
b $Q(x) = -x^2(x^2 - 4)$

c $R(x) = -x^5 + 5x^3 - 4x$

d $S(x) = \frac{1}{2}x^6 - 2x^4$

e $T(x) = x^4 + 2x^3$

f $U(x) = -x^3 + 2x^2$



ACTIVITY 1.1

1.1 End behaviour

a. Graph III
b. Graph I
c. Graph V
d. Graph II
e. Graph VI
f. Graph IV

SOLUTION

What are Real Zeros of polynomials

Recall:

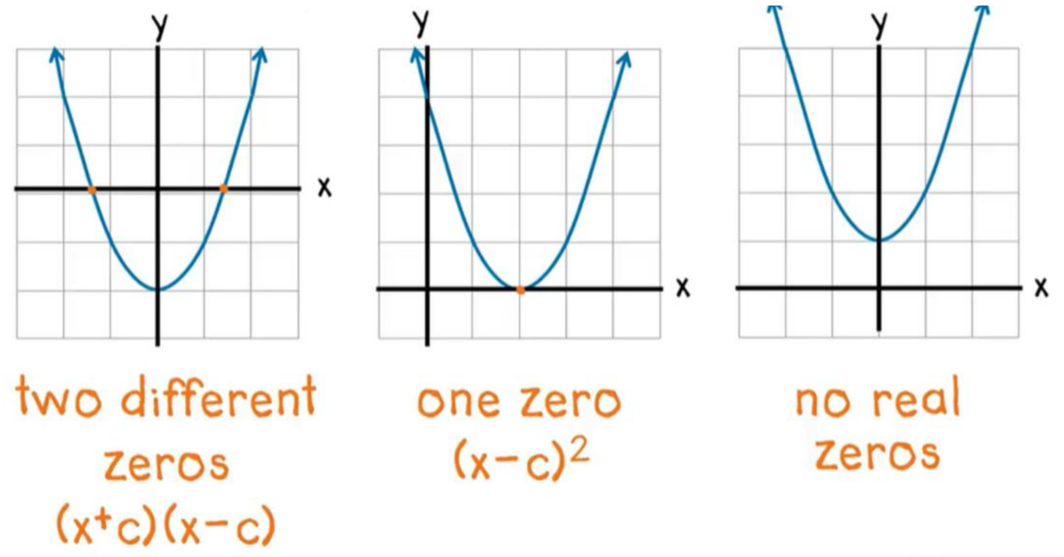
zeros of *P* are the **solutions** of the polynomial equation.

These are also known as the roots or the x-intercepts.

REAL ZEROS OF POLYNOMIALS

If P is a polynomial and c is a real number, then the following are equivalent:

1. c is a zero of P.
2. $x = c$ is a solution of the equation $P(x) = 0$.
3. $x - c$ is a factor of $P(x)$.
4. c is an x-intercept of the graph of P.



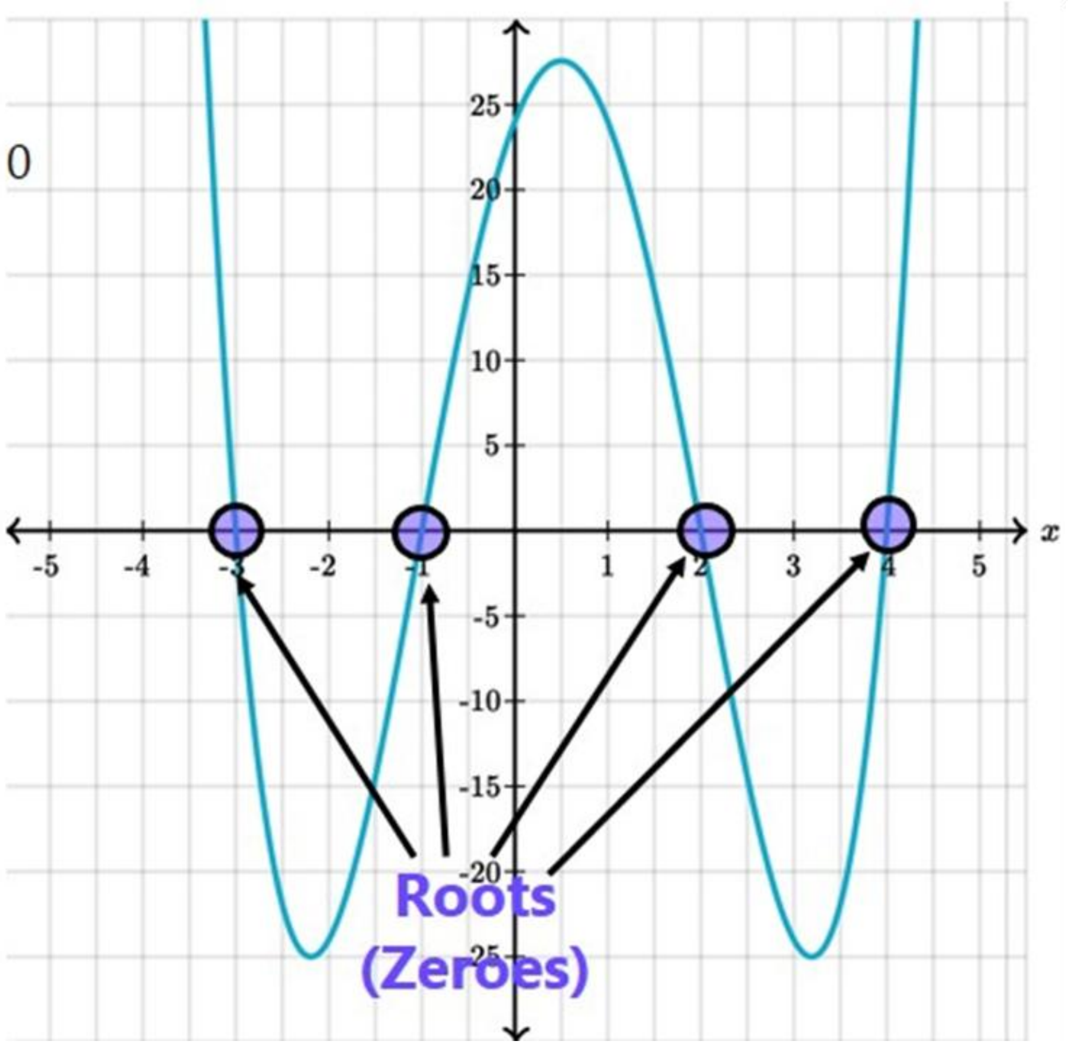
Example of Real zeros of polynomials

Zeros of Polynomial is value of x where p(x) = 0

Example

p(x) = (x + 1) (x − 2) (x + 3) (x − 4)

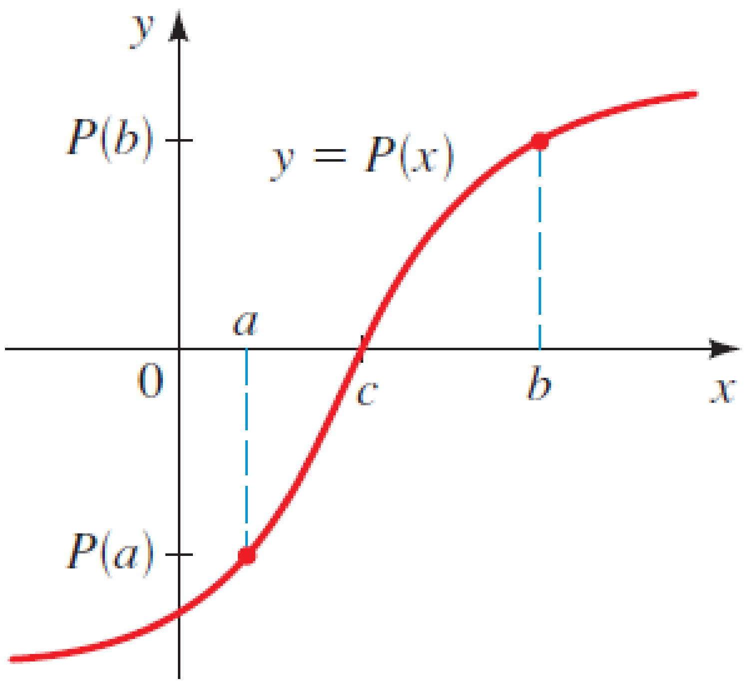
So, x = −1, 2, −3, 4 are the zeros of p(x)



Intermediate value theorem

INTERMEDIATE VALUE THEOREM FOR POLYNOMIALS

If P is a polynomial function and $P(a)$ and $P(b)$ have opposite signs, then there exists at least one value c between a and b for which $P(c) = 0$.



if a polynomial changes from positive to negative (or vice versa) between two points, **it must cross X-axis somewhere in between**.

We will use this theorem to graph polynomials.

How to graph a polynomial function

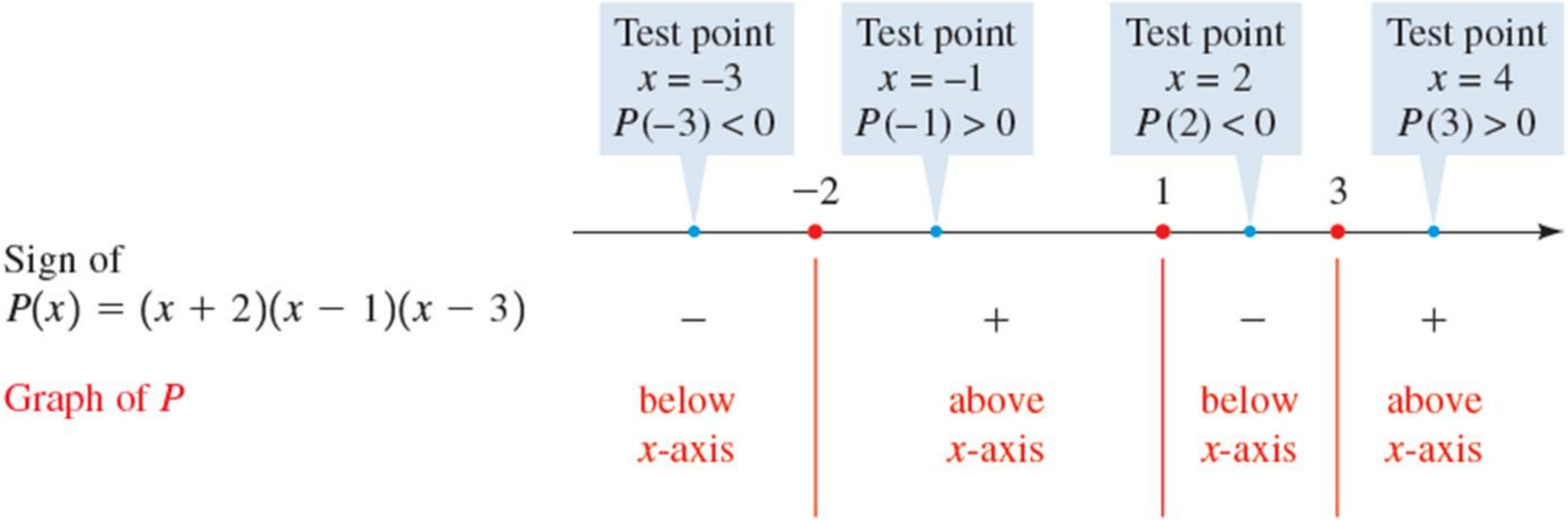
GUIDELINES FOR GRAPHING POLYNOMIAL FUNCTIONS

1. **Zeros.** Factor the polynomial to find all its real zeros; these are the x-intercepts of the graph.
2. **Test Points.** Make a table of values for the polynomial. Include test points to determine whether the graph of the polynomial lies above or below the x-axis on the intervals determined by the zeros. Include the y-intercept in the table.
3. **End Behavior.** Determine the end behavior of the polynomial.
4. **Graph.** Plot the intercepts and other points you found in the table. Sketch a smooth curve that passes through these points and exhibits the required end behavior.

Using Zeroes to Graph polynomials

Sketch the graph of the polynomial function $P(x) = (x + 2)(x - 1)(x - 3)$.

SOLUTION The zeros are $x = -2$, 1, and 3. These determine the intervals $(-\infty, -2)$, $(-2, 1)$, $(1, 3)$, and $(3, \infty)$.



We will use **test points** in intervals to determine the "**sign of the polynomial**":

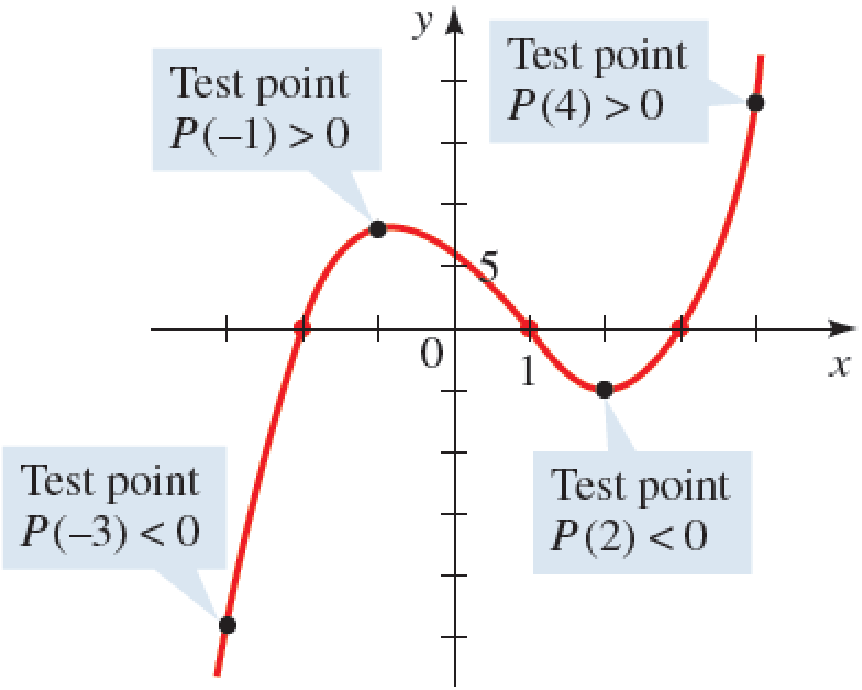
That is, whether the polynomial lies **above** or **below the x-axis** in that interval.

EXAMPLE

Using Zeroes to Graph polynomials

Plotting a few additional points and connecting them with a smooth curve helps us to complete the graph in Figure 7.

	x	$P(x)$
Test point →	−3	−24
	−2	0
Test point →	−1	8
	0	6
	1	0
Test point →	2	−4
	3	0
Test point →	4	18



EXAMPLE

Finding Zeros and Graphing a Polynomial Function

Let $P(x) = x^3 - 2x^2 - 3x$.

(a) Find the zeros of P. **(b)** Sketch a graph of P.

SOLUTION

(a) To find the zeros, we factor completely.

$$\begin{aligned} P(x) &= x^3 - 2x^2 - 3x \\ &= x(x^2 - 2x - 3) && \text{Factor } x \\ &= x(x - 3)(x + 1) && \text{Factor quadratic} \end{aligned}$$

Thus the zeros are $x = 0$, $x = 3$, and $x = -1$.

(b) The x-intercepts are $x = 0$, $x = 3$, and $x = -1$. The y-intercept is $P(0) = 0$. We make a table of values of $P(x)$, making sure that we choose test points between (and to the right and left of) successive zeros.

Since P is of odd degree and its leading coefficient is positive, it has the following end behavior:

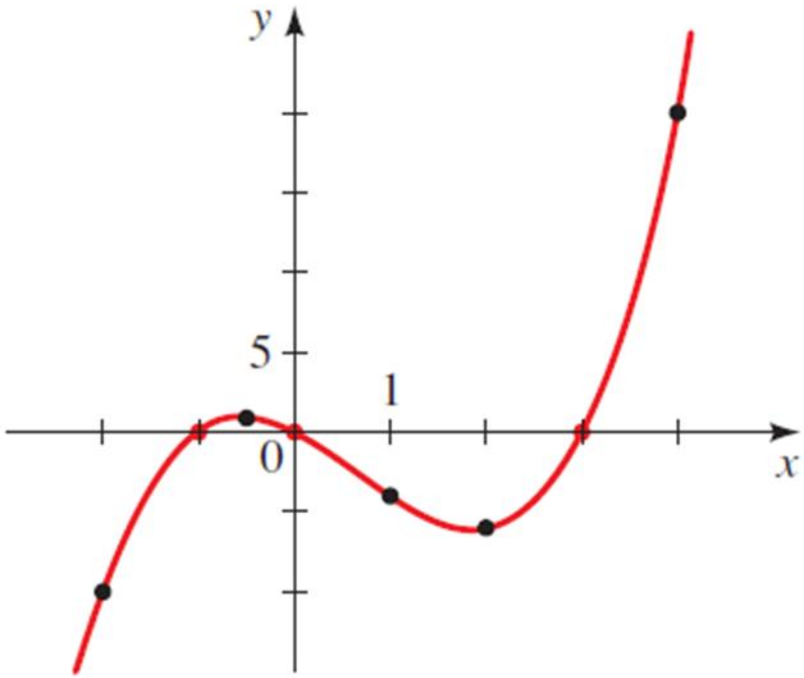
$$y \to \infty \quad \text{as} \quad x \to \infty \qquad \text{and} \qquad y \to -\infty \quad \text{as} \quad x \to -\infty$$

EXAMPLE

Activity: Finding Zeros and Graphing a Polynomial Function

We plot the points in the table and connect them by a smooth curve to complete the graph

	x	$P(x)$
Test point →	-2	-10
	-1	0
Test point →	$-\frac{1}{2}$	$\frac{7}{8}$
	0	0
Test point →	1	-4
	2	-6
	3	0
Test point →	4	20



EXAMPLE

1.2 Graphing factored polynomials

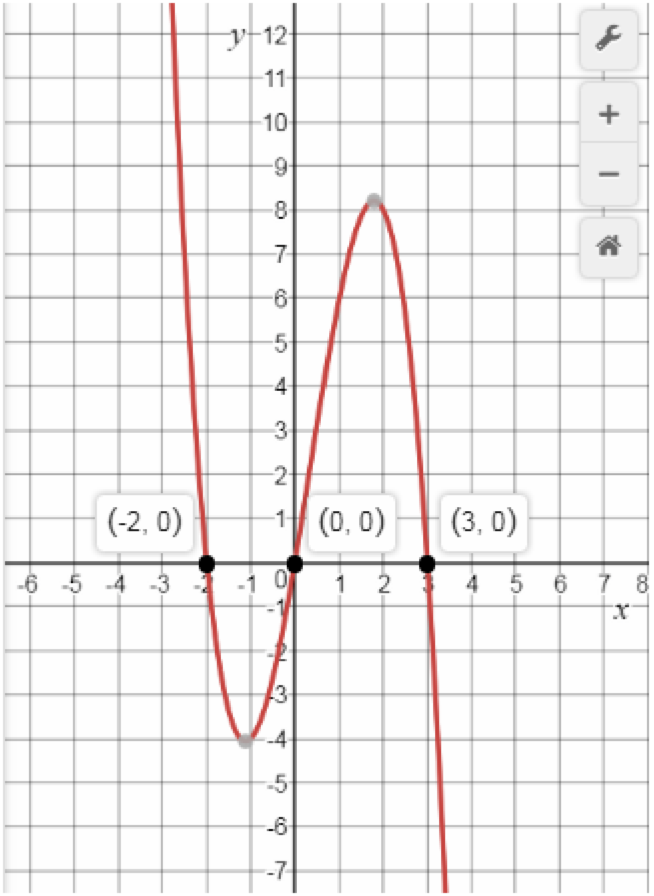
(a) $P(x) = -x(x - 3)(x + 2)$

(b) $P(x) = (x + 2)(x + 1)(x - 2)(x - 3)$

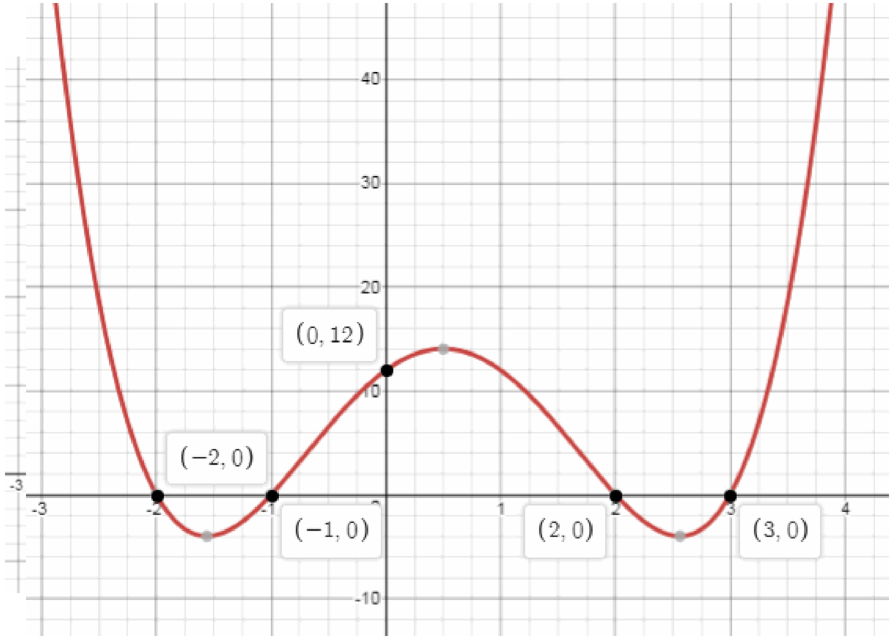
ACTIVITY 1.2

1.2 Graphing factored polynomials

(a) $P(x) = -x(x-3)(x+2)$

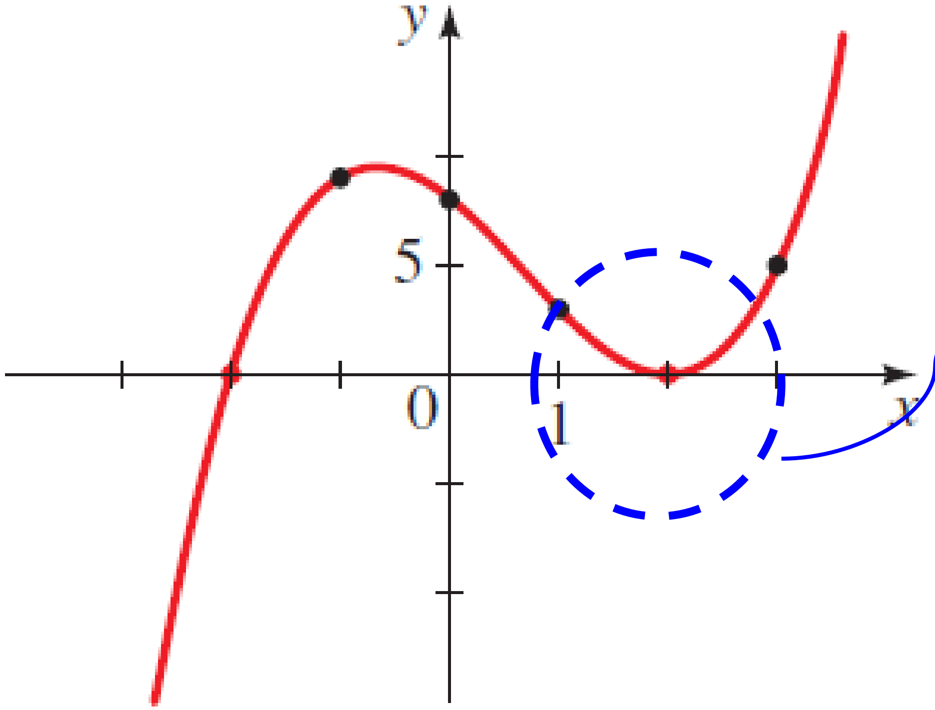


(b) $P(x) = (x+2)(x+1)(x-2)(x-3)$



SOLUTION

Multiplicities and the shape of a graph **near a zero**



Observe the graph **does not cross the x-axis** at the x-intercept 2.

This is because the factor $(x-2)^{2}$ corresponding to that zero is raised to an **even power**,
so **it doesn't change sign** on either side of 2.

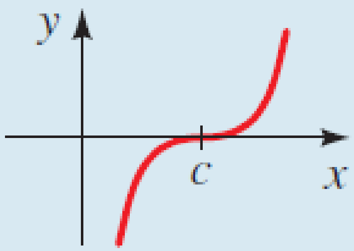
Multiplicities and the shape of a graph **near a zero**

SHAPE OF THE GRAPH NEAR A ZERO OF MULTIPLICITY *m*

If c is a zero of P of multiplicity m, then the shape of the graph of P near c is as follows.

Multiplicity of *c* | **Shape of the graph of *P* near the *x*-intercept *c***

m odd, $m > 1$



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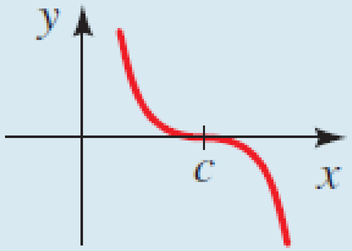
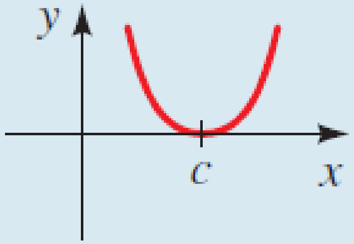


m even, $m > 1$



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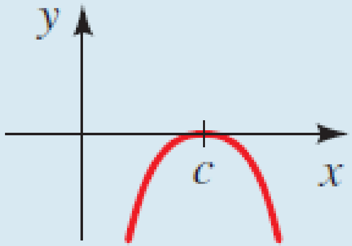


1.3 Graphing factored polynomials (with multiplicities)

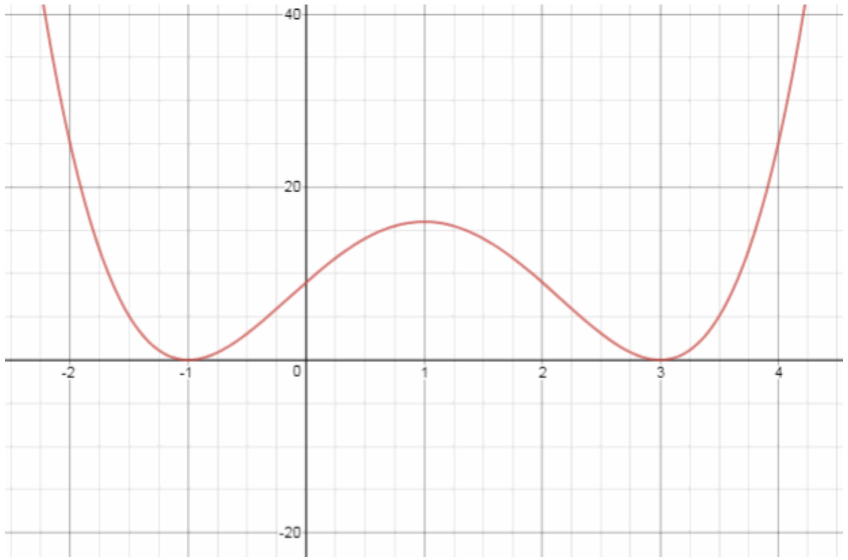
(a) $P(x) = (x - 1)^2(x + 2)^3$

(b) $P(x) = (x - 3)^2(x + 1)^2$

ACTIVITY 1.3

1.3 Graphing factored polynomials

(b) $P(x) = (x - 3)^2(x + 1)^2$



End behavior:
When $x \to -\infty$, both squared factors are positive,
$P(x)$ is positive to the far left.

When $x \to +\infty$, , both squared factors are positive,
$P(x)$ is positive to the far right.

Intercepts:
$x + 1 = 0, \qquad x - 3 = 0$
x-intercepts:
$x = -1$, double (graph touches x and turns).
$x = 3$, double (graph touches x and turns).

y-intercept: $P(0) = 9$
$P(1) = 16$, so the graph rises through $(0, 9)$

SOLUTION

1.3 Graphing factored polynomials

(a) $P(x) = (x - 1)^2(x + 2)^3$

End behavior:
When $x \to -\infty$, the squared factor is positive, the cubed is negative
$P(x)$ is negative to the far left.

When $x \to +\infty$, both factors are positive,
$P(x)$ is positive to the far right.

Intercepts:
$x - 1 = 0, \quad x + 2 = 0$
x-intercepts:
$x = -2$, triple (graph flattens out at the crossing).
$x = 1$, double (graph touches x and turns).

y-intercept: $P(0) = (-1)^2(2^3) = 8$
$P(0.5) \approx 3.9$, so the graph falls through $(0, 8)$

SOLUTION